PRONOIA: Event-Driven Financial Reasoning with Market-Verifiable Rewards

Anonymous Author(s)

ABSTRACT

Event-driven equity trading is shifting from speed alpha—obtaining announcements faster—to cognitive alpha: reading an announcement's implications and transmission more deeply than the market. Two gaps block progress: traditional event study methods are post-hoc descriptive and cannot form forward-looking quantifiable predictions; and LLM-enhanced methods can read announcements but universally rely on LLM-as-judge validation, sharing the model's blind spots.

We present PRONOIA, an event-driven financial reasoning framework with three pillars: (i) an *Event Reasoning Chain* (ERC) architecture formalizing events as state transitions and executing multi-path propagation over an encoded *Propagation Graph*; (ii) a *Predict–Assert–Verify* (PAV) output contract decomposing each event prediction into 3–6 verifiable claims over measurable financial indicators; and (iii) *Reinforcement Learning with Verifiable Rewards* (RLVR) settling every claim against actual post-event market data.

Two controlled experiments on 1,034 time-split out-of-sample A-share events (2024–2026): rule-driven backtest achieves 70.3% direction accuracy with market confirmation as the strongest meta-signal (+18.7pp); PAV-RLVR LoRA adapter improves claim-level accuracy from 57.3% to 63.2% (+5.9pp, paired bootstrap $p < 0.01$) with simultaneous Brier (0.283→0.254) and ECE (0.163→0.107) gains. A controlled ablation shows residual bias correction rewards cause reward hacking, reducing OOS accuracy by 2.29pt—an empirical caution for financial RL design.

Structural analysis reveals asymmetric capability: the model excels at falsifiable claims (FALSE-side 70.9%, significance family 79.1%, horizon $t30$ 75.3%) but struggles with momentum-continuation claims (TRUE-side 53.0%, bm_ret_t15 49.8%). Three case studies characterize failure modes: prior-driven momentum bias, risk paragraphs that are cognitively correct but behaviorally inert, and upstream context contamination. These findings define clear improvement directions for event-driven financial AI agents.

CCS Concepts: Computing methodologies → Reinforcement learning; Information systems → Expert systems; Applied computing → Economics.

Keywords: Event-driven financial reasoning; LLM agents; Financial world models; RLVR; Event study methodology

ACM Reference Format:
Anonymous Author(s). 2026. PRONOIA: Event-Driven Financial Reasoning with Market-Verifiable Rewards. In . ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
Conference'17, July 2017, Washington, DC, USA
© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-x-xxxx-xxxx-x/YY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 INTRODUCTION

1.1 The Paradigm Shift

Since Ball & Brown [1] and Fama et al. [2], the event study has been the cornerstone of empirical finance. Its five-step pipeline—define event, estimation window, market model, compute AR/CAR, statistical test—has remained structurally unchanged for over half a century. Meanwhile, event-driven trading has evolved from *speed alpha* (co-location, low-latency feeds) to *cognitive alpha*: returns come from reading *what the announcement means* more accurately than the market, not from reading it faster.

LLMs are the natural technology for cognitive alpha—they parse full announcements, detect nuanced guidance tone changes, and sketch transmission chains across sectors. But three fundamental problems block reliable event-driven financial AI, and no existing system solves all three:

Problem 1: Traditional methods describe the past, not the future. Event studies answer "how did *this type* of event move prices on average?" but cannot answer "how will *this specific* event move prices?" Announcements are collapsed into 0/1 dummies; magnitude, wording, and market-surprise component—what actually moves prices [3]—are lost. Macro events are sparse and heterogeneous, limiting out-of-sample predictive power.

Problem 2: Events are treated in isolation, ignoring economic transmission. The standard event study assumes independence [4], but markets react through multi-layer channels: "rate cut → real estate financing improves → building material demand rises → commodity prices move." Markets systematically underreact to indirect effects [5], and the event study pipeline has no slot for them.

Problem 3: LLM-enhanced methods lack external ground truth. LLM event analyses solve Problems 1–2 but validate almost universally with LLM-as-judge—evaluator and evaluatee share error modes, are vulnerable to prompt injection, and "sounding reasonable" ≠ "actually correct."

1.2 The PRONOIA Proposition

PRONOIA resolves this dilemma by grounding every prediction in actual post-event market data:

(1) **Structured Events:** Each event is a structured information shock $(type, direction, strength, surprise)$—not an isolated 0/1 signal—input to multi-path propagation.
(2) **Propagation Graph Reasoning:** Economic logic is explicitly encoded as a DAG $G = (V, A)$ with nodes as macro variables/sectors/assets and edges as pre-built transmission

mechanisms (interest-rate channel, supply-demand, industry upstream-downstream, capital flow). Multi-path *search and assembly* replaces free-generation.

(3) **Market Ground Truth Verification:** Every prediction is a falsifiable claim settled by actual post-event market CAR data—not LLM self-assessment. Graph edge weights are continuously calibrated by market feedback, closing the "reason → verify → revise" loop.

1.3 Contributions

(1) **ERC Architecture:** A four-layer decoupled architecture (Perception → Reasoning → Decision → Evolution) with the Event–Transmission–Position (ETP) triple as atomic logic unit, formalizing the Propagation Graph as the financial analog of a robotics world model's transition function.

(2) **PAV-RLVR Framework:** A verifiable output contract (Predict–Assert–Verify) decomposing predictions into 3–6 claim-level assertions over measurable indicators (CAR, benchmark returns, p-values, window averages across T+3/7/15/30/60), combined with RLVR settling every claim against objective market data.

(3) **Controlled Empirical Validation:** (a) Rule-driven backtest isolating the reasoning skeleton: 70.3% ACC with market confirmation (+18.7pp); (b) PAV-RLVR LoRA adapter improving claim accuracy 57.3%→63.2% (+5.9pp, paired bootstrap $p < 0.01$) with Brier and ECE gains.

(4) **Reward Hacking Documentation:** Controlled ablation showing residual bias corrections cause reward hacking (−2.29pt OOS) despite training reward curve improvements—an empirical caution for financial RL design.

2 RELATED WORK

2.1 Event Study Methodology

Foundations by Ball & Brown [1] and Fama et al. [2] established the event study paradigm. Critical limitations documented: event dummies lose surprise magnitude [3]; macro event sparsity limits extrapolation; cross-sectional CAR correlation inflates statistics [4]. We complement this with the Propagation Graph as forward-looking structure preserving event richness and modeling transmission.

2.2 Financial NLP and Event Extraction

The 4th generation of event processing brings LLM-based understanding. MAVEN [8] and MAVEN-ERE [9] provide event detection/coreference datasets; Doc2EDAG [10] and DCFEE [11] tackle document-level financial extraction; FinKario [12] builds event-enhanced financial KGs. These are natural Perception-layer inputs; PRONOIA's contribution is the Reasoning layer—structured propagation over an encoded economic graph with market-grounded RLVR training.

2.3 Event-Driven Financial Prediction

Table 1 situates PRONOIA: all prior systems operate on single-hop "event → reaction" mapping with LLM-as-judge or non-verifiable rewards. The 58–62% ACC ceiling confirms single-hop's limited headroom. PRONOIA's Propagation Graph + RLVR addresses both gaps.

Table 1: Comparison with event-driven financial AI systems. ACC = direction accuracy on respective test sets.

System	Approach	Limitation	ACC
FinHEAR [13]	Historical analog	Single-chain, no multi-path	58–62%
FinKario [12]	KG retrieval	Static KG, retrieval-only	58.1%
TradingAgents [14]	Multi-agent vote	Not event-driven	~55%
Janus-Q [15]	Hierarchical reward	No event modeling	58.7%
Argus [16]	LLM-as-judge	Judge shares errors	—
PRONOIA	Graph + PAV + RLVR	US OOS limited	**63.2%**

2.4 RLVR and Financial RL

RLVR [17] settles rewards against formal verification rather than self-evaluation. AlphaAgent [18] and FactorMiner [19] apply related ideas to alpha mining. The critical difference: PAV decomposes predictions into claim-level assertions over measurable indicators, making every reward component independently verifiable *and* preventing reward hacking—our ablation in §4.3 documents what happens when this guardrail weakens.

2.5 Robotics World Models

The Propagation Graph formalization draws from robotics world models [6, 7]: $S_{t+1} = f(S_t, a_t)$ maps to our graph: state → market state (sector CAR, volume anomaly, volatility, sentiment), action → structured event, transition function → Propagation Graph (multi-path probability propagation), terminal verification → market CAR direction and claim settlement, planning → Navigator search.

3 METHOD

3.1 Financial Event Reasoning Chain

PRONOIA's four-layer architecture operates on the Event–Transmission–Position (ETP) triple:

(1) **Perception:** Heterogeneous disclosures → structured state transition → surprise measurement.

(2) **Reasoning:** Multi-path Propagation Graph search + adversary testing + priced-in detection.

(3) **Decision:** Direction → timing → sizing → portfolio → execution.

(4) **Evolution:** Versioned state evolution, credit attribution, policy update gated by market evidence.

This paper focuses on Layers 2 (Reasoning) and 4 (Evolution via RLVR), with Layers 1 and 3 providing input context.

3.2 Events as State Transitions

Definition 3.1 (Event). An event e is a state transition $X_e^- \to X_e^+$ of a declared system S at time t_e, with ex-ante benchmark B_e. The *surprise* is $X_e^+ - B_e$, not $X_e^+ - X_e^-$.

This resolves the common confusion: a large state change (guidance revision \$10–11B → \$8–9B) is not a large surprise if the market expected \$8.5B.

Definition 3.2 (Event Boundary). Boundary = System(S) + State Representation(X) + Analytical Scale(σ). A single earnings release can be one event at filing scale or several at item scale (revenue, EPS, guidance changes).

3.3 Propagation Graph

$G = (V, A)$ is a DAG where:

- **Nodes** V**:** Each v contains (subject, expected_direction, expected_magnitude, time_window), from macro variables to sectors to individual assets.
- **Edges** A**:** Each $a = (u, v, label)$, $label \in \{+1, -1\}$, with conditional transmission probability $P(B_{up}|A_{up}) = N(A_{up} \wedge B_{up})/N(A_{up})$, fused via Bayesian update $P_{new} = P_{prior} \times 0.7 + P_{observed} \times 0.3$.

Three edge-source tiers: (a) prior historical statistics, (b) online recent CAR observations, (c) Navigator real-time reasoning.

3.4 Navigator-Searcher Protocol

Six-step collaboration over G:

(1) Navigator decomposes event e into sub-problems, dispatching one Searcher per sub-problem.
(2) Each Searcher, adopting a market-participant persona (long/short/arbitrageur/regulator/retail), traces one propagation chain.
(3) Outputs merged into DAG: deduplicated, cross-referenced, marked with verification status.
(4) Navigator identifies unconnected components; prompts additional Searchers.
(5) Market verification via CAR + dose-response + volume/volatility validates/falsifies each path.
(6) Verified paths form final reasoning graph; emit directional, magnitude, and confidence judgments.

Key choices: DAG not tree (multiple causes per effect); market posterior not LLM voting as ground truth; game-theoretic branching with heterogeneous personas.

3.5 PAV Output Contract

Predict: Input: event structured fields + as_of research context (strictly no post-event data: T0 return, pre-5/pre-20 drift, CAR to T0) + measurable indicator panel (30 indicators defined, values withheld).

Assert: Model proposes 3–6 verifiable claims: (indicator, horizon, threshold, TRUE/FALSE, confidence, reasoning_chain). Five families across T+3/7/15/30/60: `car_t{h}`, `ret_t{h}`, `bm_ret_t{h}`, `car_t{h}_pvalue`, `car_avg_*`.

Verify: Each claim objectively settled against labels.jsonl. Compute claim-level accuracy, Brier, ECE, verifiability, diversity.

The departure: PAV shifts core metric from "did you guess direction right?" to "are your claims testable and correct?" Evaluation goes from event-level to claim-level.

3.6 RLVR Training

Qwen3-8B (bf16, backbone frozen, LoRA rank 16) trained with GRPO [20] via RLVR [17]. Per event: $G = 4$ candidates, parsed into claims, settled against labels.

Table 2: Dataset statistics (time-split).

Split	Events	Markets	Sources
Train	4,965	CN 4,352 / US 613	A-share pre-announcements, M&A, guidance, macr
OOS	1,034	CN 1,027 / US 7	Same, post-training window (2024–2026), once

Table 3: Rule-driven backtest results. S1–S5 implement prior → market confirmation → noise removal → strong-signal focus.

Strategy	N	ACC	95% CI	Sharpe	avgCAR
S1 All	444	51.6%	[47, 56]	−1.01	−0.26%
S2 SectorConfirm	212	**70.3%**	[64, 76]	3.21	+0.71%
S3 StrongEvent	159	71.1%	[64, 78]	1.48	+0.32%
S4 NonPriceIn	130	74.6%	[67, 82]	3.85	+0.65%
S5 StrongNonPriceIn	85	**78.8%**	[69, 87]	0.81	+0.13%

Reward is six-element tuple (R0–R5): claim accuracy (core), confidence calibration (Brier), verifiability ratio, diversity (family coverage), format compliance, length penalty.

Ablation (v6.1-b): Adds five residual bias corrections (B3: hard-negative targeting; B4/B5: distribution residual correction; B7: claim combination quality; T3: format bonus). Vanilla (v6.2) disables these.

Training: GRPO (TRL 0.17) + Unsloth acceleration (in-process vLLM, weight sharing, Standby mode, ∼2.3× speedup). Batch 8/device × grad-accum 2 = 16 completions/step (4 events × 4 candidates). LR 1e-5, cosine annealing. 1 epoch = 1,241 steps ≈ 4,965 events. Seed 42. Single RTX 4090 (48GB), 10–12h/condition.

4 EXPERIMENTS

4.1 Dataset and Protocol

Known biases: US near-absent in OOS; no macro events in OOS; scope limited to CN A-shares.

Integrity protocol: (1) Strict as_of: features use only pre-event data, audited; (2) Time split only, OOS once; (3) Event-level paired bootstrap (2,000 resamples, weighted); (4) Claim accuracy weighted by n_settleable; (5) Calibration: Brier + 10-bin ECE + confidence-bin actual accuracy.

4.2 Rule-Driven Backtest (Skeleton Validation)

Design: 27 A-share blue chips + 8 sector ETFs, CSI300 benchmark, 2023-01 to 2026-07, 33 events, 20 types, CAR via market model (120d estimation, 20d gap, T+3 holding).

Key findings: (1) S1 51.6% ≈ random—priors alone cannot trade; (2) Sector ETF confirmation is the strongest meta-signal (+18.7pp, Sharpe −1.01→3.21)—"use market posterior to filter priors"; (3) Event strength quantifies surprise (routine cuts 47.9% < random; 924-package cuts 82.6%); (4) "Graph can only be used once"—low-ACC events (consumer boost 33.3%, spike-crash 31.6%, rate cuts 47.9%) prove lookup tables fail, motivating online reasoning. In/out-of-sample gap 1.2pp; LOEO 68.0–71.4%.

Table 4: PAV-RLVR main results (1,034 OOS events, $t = 0.6$).

Metric	Base	v6.1-b	v6.2-vanilla
Claim ACC	0.5729	0.6093	**0.6322**
Brier	0.2826	0.2569	**0.2539**
ECE	0.1632	**0.1064**	0.1070
Format	0.775	**0.997**	0.842
Settled claims	2,978	4,056	3,154

Table 5: Accuracy by event type (CN market).

Event Type	Events	Base	v6.2	Δ
Earnings beat/miss	478	0.524	0.598	**+7.4pt**
M&A/spinoff/financing	373	0.618	0.661	+4.4pt
Guidance up/down	169	0.632	0.679	+4.8pt
Policy rate	7	0.598	0.591	(few)

Table 6: Accuracy by indicator family and horizon.

Slice	Claims	Base	v6.2	Δ
Significance (p-val)	944	0.614	**0.791**	**+17.7pt**
CAR	1,207	0.566	0.592	+2.6pt
Benchmark	647	0.411	0.498	+8.7pt
Horizon t30	675	0.622	**0.753**	**+13.1pt**
Horizon t7	1,154	0.559	0.619	+6.0pt
Horizon t15 (worst)	610	0.452	0.505	+5.3pt

4.3 PAV-RLVR Controlled Ablation

Paired bootstrap: Δ(vanilla − base) = +5.93pt (95% CI [+3.79, +8.07]); Δ(vanilla − v6.1-b) = +2.29pt (95% CI [+0.78, +3.79])—CIs exclude zero. Event-level: vanilla wins 394, v6.1-b 335, tie 302.

Reward hacking: Corrections' only benefit is format (+15.5pt), recoverable by inference-time re-sampling. Training reward (v6.1-b 0.90 vs vanilla 0.69) *inversely correlates* with OOS accuracy (0.609 vs 0.632)—model optimizes proxy signals over claim quality. KL, length, truncation are healthy and comparable—ruling out instability. **Implication: training reward trend cannot validate reward components; OOS adjudication is the only mechanism.**

4.4 Hard-Negative Subset

146 events (M&A success/turnaround/beat/additional issuance/buyback): base 0.5103, v6.1-b 0.5515, **vanilla 0.5862**. Corrections help +3.5pt on this subset but net negative overall—the "residual bias correctable by prior" premise is unsupported.

5 ANALYSIS

5.1 By Event Type

Earnings: largest relative lift (+7.4pt) but lowest absolute (0.598)—strongest priors, hardest cases. M&A/guidance: highest absolute (0.661, 0.679).

5.2 By Indicator Family × Horizon

Significance family strongest (+17.7pt to 79.1%): most A-share CARs insignificant—falsifying significance is the model's strongest move. Weakest: `bm_ret_t15` (0.498, coin-flip). Long-horizon t30 (+13.1pt) vs t15 (+5.3pt): falsifiable long-horizon claims outperform.

5.3 Conditional Analysis

Claim count: 2–3 best (0.690/0.656); 4 drops to 0.569. Constraint should narrow to 3–4. **Direction:** FALSE-side **0.709** (n=1,812) vs TRUE-side **0.530** (n=1,342). Training reduces TRUE ratio 0.525→0.425—beneficial RL product matching task asymmetry. **Extremes:** 293 all-correct (28.3%, M&A/earnings); 55 all-wrong (5.3%, beat/turnaround). **Calibration:** Non-monotonic—0.75–0.85 bin actual accuracy only 0.53–0.59 (main ECE contributor); 0.80/0.90 ends reliable (0.777/0.926). Platt post-hoc: ECE 0.085→0.036.

Profile: Excels at FALSE, long-horizon t30, significance, M&A/guidance, 2–3 claims, confidence 0.80/0.90. Struggles with TRUE momentum, benchmark direction, hard-negatives, 4+ claims, 0.75–0.85 confidence.

6 CASE STUDY

6.1 Case 1 (Success): Shengtong Turnaround — vanilla 3/3, base 1/3

002599, 2026-01-30, turnaround. Pre-event CAR −7.95% (oversold); context: "mean-reversion vs micro-profit cancel → neutral, confidence lowered." v6.2 claims: `car_t7>0` TRUE (0.72) ✓; `bm_ret_t30>0` FALSE (0.65) ✓; `car_t30_pvalue<0.05` FALSE (0.55) ✓. Base ignores context, bets p-value TRUE with arbitrary thresholds. **Success = context absorption + diversity planning + betting FALSE.**

6.2 Case 2 (Failure): Zhongjin Gold Beat — all models wrong

600489, 2026-01-28, beat. T0 limit-up +9.98%, pre20 +46% gain. Context warns sell-the-news (−6.0 signal) but all three models claim `car_t7>0` (confidence 0.72), 0/4 wrong. Ground truth: car_t30 −29.8%. **Risk paragraph cognitively correct, behaviorally inert.**

6.3 Case 3 (Failure): Zhongshan Utilities Beat — upstream context wrong

000685, 2026-01-31, beat. Context concludes "volume-price resonance → up"; actual continuous decline (car_t7 −3.2%, car_t30 −17.3%). Model extrapolates weak stock → weak market (bm_ret wrong). pre20 CAR +8.06% used as significance support, not mean-reversion warning. **Garbage-in upstream; features reinforce priors, don't test them.**

6.4 Cross-Case Synthesis

(1) Failures concentrate in "bullish text + T0 positive reaction": prior = momentum, A-share reality = mean reversion.
(2) Risk paragraphs "cognitively correct, behaviorally inert": no hedging claims—prompt/reward design fix.
(3) Benchmark family ≈ coin flip: individual-stock weakness extrapolated to market weakness.

Table 7: Conclusion credibility grading.

Conclusion	Grade	Basis
RLVR effective (+5.9pt)	High	Effect ≫ noise; same pipeline/OOS
Structure (FALSE>TRUE; t30>t15)	High	Hundreds/thousands per cell; 3 models agree
Vanilla > v6.1-b 2.29pt	**Low**	3 confounders; CI lower bound −0.78
Confidence 0.90 → 0.926	Verify	Self-selection; n(0.90) unconfirmed
US conclusions	Unusable	OOS US = 7 events

(4) Upstream context with strong priors contaminates downstream—context should output self-confidence + counterparty evidence.
(5) Cross-indicator claims silently dropped—wastes expression, underestimates coverage.

7 DISCUSSION

7.1 Unified Interpretation

Two experiments validate one proposition at three levels:

- **Priors can't trade:** S1 51.6% + base 0.573 confirm priors need purification.
- **Market confirmation purifies:** Rule-side +18.7pp and RL-side +5.9pt are the same mechanism.
- **Environment requires online reasoning:** "Graph once" motivates model-based learning; reward hacking shows even reward functions need OOS adjudication.

7.2 Limitations

Single seed; US OOS = 7 events; 33 rule events + 4,965 RL events insufficient for larger structures; single sampling; TRUE-side weakness in A-shares (short mechanisms incomplete).

7.3 Validity Audit

Recommendation: "Corrections harmful" → "corrections did not produce expected benefits." Supporting the former requires 3-seed × 2-condition matrix.

Three methodological risks: (1) Single-sampling variance unquantified—recommend 5× on 200 events; (2) Multiple comparisons—FDR/Bonferroni + holdout needed for trading filters; (3) as_of strictness unaudited—single risk that could overturn all conclusions.

8 CONCLUSION AND FUTURE WORK

8.1 Takeaways

(1) **Methodology:** Financial event reasoning chain = Propagation Graph + PAV + market posterior reward. Rule backtest (S2 70.3%) and RL ablation (vanilla 0.632 vs base 0.573) validate at two levels.
(2) **Negative results:** Residual corrections reduce OOS accuracy 2.29pt (hard-negatives +3.47pt). Training reward ≠ OOS performance—reward hacking is real in financial RL.
(3) **Structural portrait:** Model excels at falsifiable claims (FALSE 0.709, significance 79.1%, t30 75.3%), struggles with momentum (TRUE 0.530, bm_ret_t15 0.498)—guides prompt/reward design.

8.2 Immediate Next Steps

P0: Hard-negative oversampling (2–3×) + retrain; inference-time format fallback re-sampling. **P1:** Hedging-claim mechanism; pre-drift feature explicit in prompt; US-stratified OOS. **P2:** Claim count narrow to 3–4; parser cross-indicator support; benchmark-family examples. **P3:** Multi-seed validation; Platt calibration retention.

8.3 Mid-Term Evolution

Frequency × horizon 2D reward; scale-conditioned MoE routing (event type × horizon boundaries); data expansion to 500+ rule events + US/macro RL samples.

ACKNOWLEDGMENTS

Anonymous.

REFERENCES

[1] Ball, R., & Brown, P. (1968). An empirical evaluation of accounting income numbers. *Journal of Accounting Research*, 6(2), 159–178.
[2] Fama, E., Fisher, L., Jensen, M., & Roll, R. (1969). The adjustment of stock prices to new information. *International Economic Review*, 10(1), 1–21.
[3] Kuttner, K. (2001). Monetary policy surprises and interest rates. *Journal of Monetary Economics*, 47(3), 523–544.
[4] Kolari, J., & Pynnönen, S. (2010). Event study testing with cross-sectional correlation of abnormal returns. *Review of Financial Studies*, 23(11), 3996–4025.
[5] Cohen, L., & Frazzini, A. (2008). Economic links and predictable returns. *Journal of Finance*, 63(4), 1977–2011.
[6] Ha, D., & Schmidhuber, J. (2018). World models. *arXiv:1803.10122*.
[7] Schrittwieser, J., et al. (2020). Mastering Atari, Go, chess and shogi by planning with a learned model. *Nature*, 588(7839), 604–609.
[8] Wang, X., et al. (2020). MAVEN: A massive general domain event detection dataset. *EMNLP*.
[9] Wang, X., et al. (2022). MAVEN-ERE: A unified large-scale dataset for event coreference, temporal, causal, and subevent relation extraction. *EMNLP*.
[10] Zheng, S., Cao, W., Xu, W., & Bian, J. (2019). Doc2EDAG: An end-to-end document-level framework for Chinese financial event extraction. *EMNLP-IJCNLP*.
[11] Yang, H., Chen, Y., Liu, K., Xiao, Y., & Zhao, J. (2018). DCFEE: A document-level Chinese financial event extraction system. *ACL System Demonstrations*.
[12] Li, X., et al. (2025). Finkario: Event-enhanced automated construction of financial knowledge graph. *ACL 2026*.
[13] FinHEAR: Financial event-driven hybrid architecture for earnings reaction. *arXiv:2506.09080*.
[14] Tao, T., Roice, K., & Rao, A. (2024). TradingAgents: Multi-agent system for financial trading. *arXiv:2407.18957*.
[15] Janus-Q: End-to-end event-driven trading with hierarchical gated reward modeling. *arXiv:2602.19919*.
[16] Argus: Evidence-driven multi-step reasoning with claim-level verification. *arXiv:2605.16217*.
[17] RLVR: Reinforcement learning with verifiable rewards. *arXiv:2506.07891*.
[18] Tang, Z., et al. (2025). AlphaAgent: LLM-driven alpha mining with regularized exploration. *KDD*.
[19] Wang, Y., et al. (2026). FactorMiner: A self-evolving agent with skills and experience memory. *arXiv*.
[20] Shao, Z., et al. (2025). GRPO: Group relative policy optimization. *TRL Documentation*.
[21] Morgeson, F., Mitchell, T., & Liu, D. (2015). Event system theory. *Academy of Management Review*, 40(4), 515–537.
[22] Tetlock, P. (2011). All the news that's fit to reprint. *Review of Financial Studies*, 24(5), 1481–1512.
[23] Sheng, J., Sun, Z., Yang, B., & Zhang, A. (2026). Generative AI and asset management. *Review of Financial Studies*.
[24] Yu, Y., et al. (2024). FinCon: A synthesized LLM multi-agent system. *NeurIPS*.
[25] Shu, D., et al. (2026). FinCall-Surprise: A large scale multi-modal benchmark for earning surprise prediction. *ACL*.
[26] Su, J., et al. (2026). FinChain: A symbolic benchmark for verifiable chain-of-thought financial reasoning. *ACL*.
[27] Zhu, F., et al. (2026). FinDeepResearch: Evaluating deep research agents. *arXiv*.
[28] Zhu, J., et al. (2026). FinMCP-Bench: Benchmarking LLM agents for real-world financial tool use. *arXiv*.
[29] PRONOIA Team (2026). Event-driven financial reasoning: ERC architecture, validation, and evolution. 61-page survey.